

Candidate Domain Concepts

Candidate Domain Concepts — Extraction & Classification

University of Botswana Service Queue and Appointment Management System — CSI473 Lab 04, Studio Step 1 (00-18 min) **Source:** Laboratory 3, Group 17 (functional requirements FR-01-FR-10, actor goals, UC-01 “Join Virtual Queue,” acceptance criteria, use case diagram)

This worksheet does two things the brief asks for in the first studio block: 1. **Extracts** every candidate noun, role, event, and rule mentioned or implied in the Lab 03 artefacts. 2. **Classifies** each one, and explicitly **rejects** the terms that describe the interface or the technology rather than the problem domain, so they don’t leak into the domain model in the 18-50 min block.

1. Candidate concepts, by domain-modelling category

Categories follow the standard “list of category names” technique (things, roles, events, transactions, rules, containers) so nothing is missed and nothing is admitted just because it’s a noun.

#	Candidate term	Where it comes from	Category	Verdict	Rationale
1	Student	Actor; FR-01-08	Role	✓ Keep	Primary actor who initiates queueing and appointment behaviour — has domain responsibilities (join, cancel, hold a queue position).
2	Registration Staff	Actor; FR-09-10	Role	✓ Keep	Acts on queues/appointments; distinct responsibilities from Administrator.
3	Registration/Service Administrator	Actor; diagram	Role	✓ Keep	Configures services and appointment availability — a distinct role from Staff.
4	Registration Service	FR-01, FR-05; “Select Registration Service”	Concept (description)	✓ Keep	The thing being queued for/booked against (e.g. transcript requests, ID replacement). Core domain concept, not a UI screen.
5	Virtual Queue	FR-02; UC-01	Concept (container)	✓ Keep	Holds an ordered set of Queue Entries for one service; has a capacity and open/closed state.
6	Queue Entry	UC-01 main flow step 5; postconditions	Concept (transaction)	✓ Keep	The record created when a student joins — distinct from the Student and from the Queue itself.
7	Queue Number	FR-03; UC-01 step 6	Concept (value/identifier)	✓ Keep	A unique identifier assigned to a Queue Entry — a domain value, not just a UI label.
8	Queue Position	FR-04; UC-01 step 7	Concept (value, derived)	✓ Keep	Calculated rank of a Queue Entry within a Virtual Queue — worth an attribute/derived value, not a screen field.
9	Queue Status	FR-04; UC-01 step 9	Concept (value/state)	✓ Keep	State of a Queue Entry (e.g., waiting, called, completed) — needed for the state-based business rules in block 50-70.
10	Appointment Slot	FR-05-07	Concept (thing)	✓ Keep	A bookable time slot for a service — exists independently of any booking.
11	Appointment	FR-06, FR-08; diagram	Concept (transaction)	✓ Keep	The booking itself, linking a Student to an Appointment Slot — distinct from the Slot.

12	Cancellation Rule / Cancellation Policy	FR-08; "Validate Cancellation Policy"	Rule/Policy	✓ Keep	Named business policy governing when an Appointment can be cancelled — becomes one of the ≥6 business rules.
13	Eligibility (student eligibility for a queue/service)	UC-01 step 4; "Call Next Eligible Student"	Rule/Policy	✓ Keep	A named rule concept, not a UI check — governs who may join a queue or be called next.
14	Service Rule (call-next rule)	FR-10	Rule/Policy	✓ Keep	Distinct policy from the cancellation rule — decides call order (e.g., FIFO, priority). Needs its own name once the team defines it.
15	Notification (queue/appointment update)	Diagram: "Receive Notifications"	Concept (event/message), borderline	⚠ Keep, narrowly	Keep the <i>business event</i> "student is notified of a status change." Drop the delivery mechanism (SMS/email/push) — that's technology, see §2.
16	Report / Activity Record	"View Reports and Activity"	Concept, borderline	⚠ Defer	Only keep if the team's requirements actually define a persisted report entity. As written it reads as a UI/query feature, not a domain thing — revisit once revised requirements are on hand.
17	One-active-entry-per-service rule	UC-01 "BUSINESS RULE"	Rule/Policy	✓ Keep	"A student cannot have more than one active queue entry for the same registration service" — an invariant, feeds block 50-70 directly.
18	Join Virtual Queue (event)	UC-01	Event	✓ Keep	Noteworthy domain event — creates a Queue Entry.
19	Book Appointment (event)	FR-06	Event	✓ Keep	Creates an Appointment against a Slot.
20	Cancel Appointment (event)	FR-08	Event	✓ Keep	Governed by Cancellation Rule.
21	Call Next Eligible Student (event)	FR-10	Event	✓ Keep	Changes Queue Entry status; governed by the Service Rule.
22	Complete Student Service (event)	Diagram	Event	✓ Keep	Closes out a Queue Entry/Appointment.

2. Terms rejected as interface or technology terms

These appeared in the Lab 03 wording but describe **the software's interface, an action a user performs on a screen, or an implementation mechanism** — not a problem-domain idea — so they're deliberately left out of the domain model, per the brief's own minimum standard ("not merely a screen, controller, framework type or database table").

Rejected term	Where it appears	Why it's excluded
System	Every FR ("the system shall...")	The application itself — a boundary/actor in the use-case sense, not a domain class.
Select (as in "Select Registration Service")	FR-01, actor goal	A UI interaction verb. The underlying domain concept it points to — <i>Registration Service</i> — is kept; the click/tap isn't.
Display / View (as in "system displays...", "View Queues", "View Appointments", "View Appointment Details")	FR-04, FR-09, diagram	Presentation behaviour — showing data on a screen — not a domain responsibility. The underlying data (Queue Position, Appointment) is already captured above.
Screen / Screenshot	Implied by "display" language; explicitly ruled out in the Lab 04 brief itself	Interface artefact, not a domain concept.
Login / access the system	UC-01 precondition 1 ("The student can access the system")	Authentication/authorisation mechanism — infrastructure, not domain.

Button / menu / form field (implied by “selects”, “chooses”)	UC-01 flow	UI widget vocabulary, not present as nouns but worth flagging so the team doesn’t introduce them while diagramming.
Notification delivery channel (SMS/email/push)	Not stated explicitly, but implied by “Receive Notifications”	If the team’s revised requirements specify a channel, that’s a technology/integration detail — keep the <i>event</i> (student notified), drop the <i>transport</i> .

3. Notes for the next studio block (18-50 min)

- You now have enough kept concepts to justify **associations**: Student-Virtual Queue (via Queue Entry), Virtual Queue-Registration Service, Student-Appointment-Appointment Slot, Registration Service-Appointment Slot.
- Two rows are flagged **borderline** (#15 Notification, #16 Report). Decide on those with your teammates *before* modelling — that decision itself is a good candidate for one of the two required “rationale for a modelling choice” write-ups, or for `decisions/D-002.md`.
- Composition candidates to test against the “lifetime ownership” rule the brief calls out: does a Virtual Queue *own* its Queue Entries (i.e., does a Queue Entry cease to exist if the Queue is deleted)? Worth a deliberate call rather than a default.
- You’ll likely want to name the call-order policy (#14) explicitly — “FIFO” vs. “priority-based” — since right now it’s only implied by FR-10 and the diagram’s “Call Next Eligible Student,” not stated as a rule.